WELLINGTON TATENDA GWAVAVA

SOC Analyst | Detection Engineer | AI Security Researcher | PhD Candidate, Cyber Studies

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PROFESSIONAL SUMMARY

SOC analyst and detection engineer completing a PhD in Cyber Studies at the University of Tulsa, with operational SOC experience and published research on SOC goal incongruency and goal-metric mapping. Deployed the full CISA Logging Made Easy (LME) stack in a live university SOC, where it was adopted as the operational standard. Builds agentic AI systems for security work, from a multi-model LLM research pipeline to a multi-agent platform shipped in 22 hours at Hacklahoma 2026. Holds ISC2 Certified in Cybersecurity (CC), the Google Cybersecurity and Google AI Professional Certificates, and Cribl Certified Admin Level 2; Secretary, ISC2 Tulsa Metro Chapter. My working pattern is to build the system, observe what it actually does, optimize for desired outcomes.

CORE COMPETENCIES

Security Operations and SIEM: CISA LME (full stack operational deployment); Elastic Stack ELK; Cribl Stream (Certified Admin L2, pipeline configuration); Wazuh HIDS/EDR; SentinelOne (hands-on); Varonis (hands-on); Splunk (workshop-level: investigation searches, SPL); MITRE ATT&CK; MITRE D3FEND; NIST CSF; CIS Controls

Detection and Incident Response: Detection engineering and alerting workflows; multi-source log integration, enrichment, and normalization; incident triage and containment; threat hunting; endpoint alert monitoring and investigation; security metrics design and SOC governance; OSINT methodology and breach analysis

AI Security and Agentic Systems: Multi-agent AI architecture (Claude API); local LLM deployment (Ollama, DeepSeek-R1, Llama); LangChain (workshop-level, NVIDIA DLI: LCEL chains, structured outputs, tool-using agents); MCP (Model Context Protocol); LLM-assisted semantic analysis with human-in-the-loop validation; agentic workflow analysis and failure mode characterization; AI feedback loop design; prompt engineering for security use cases; audit trail and state verification design for agentic actions

Engineering and Automation: Python (log ingestion and security reporting automation); TypeScript, JavaScript, React, TanStack Start, Node.js; Cloudflare Workers; Docker; Git; SQL; AWS Cloud; Linux; Windows Server; TCP/IP

PROFESSIONAL EXPERIENCE

SOC Research Externship (Part-time) | *The University of Tulsa* Aug 2025 - Present

Tulsa, OK | Applied operational SOC fieldwork

- Investigate security events in a live university SOC and evaluate how well AI-augmented detection workflows support alert triage; track US-CERT and vendor advisories for situational awareness.
- Apply doctoral research on CISO-analyst strategic alignment in a live SOC environment.

Security Operations Center Analyst | *The University of Tulsa* May 2025 - Aug 2025

Tulsa, OK | Full-time SOC internship

- Delivered the full CISA LME stack in an operational SOC; the deployment was adopted as the SOC's operational standard. Wrote Python automation for multi-source log ingestion and configured Cribl Stream pipelines for routing, enrichment, and normalization across Windows and Linux endpoints.
- Built custom Elastic Stack dashboards for lateral movement and credential dumping detection, fed by Wazuh HIDS/EDR and Elastic Agent endpoint telemetry; worked with SentinelOne and Varonis for endpoint and data security coverage.
- Validated detection rules against MITRE ATT&CK and MITRE D3FEND; set success criteria before deployment, automated repeatable security reporting in Python, and presented quantified findings to the CISO and SOC leadership.
- Authored deployment documentation, including installation guides, field dictionaries, and operational runbooks.

Graduate Research Assistant and TU Cyber Fellow | *The University of Tulsa* Aug 2023 - Present
Tulsa, OK

- Doctoral research on SOC goal incongruency, designing quantitative metrics that separate risk mitigated from activity volume. Dissertation adviser: Tyler Moore. Expected completion: May 2027.
- Reached greater than 85 percent cross-model agreement with SOC-Job-Task-Analyzer, an agentic LLM pipeline that extracts and classifies SOC analyst task duties from 49 job descriptions, with human-in-the-loop validation. GitHub: github.com/wellytg/SOC-Job-Task-Analyzer
- Published two peer-reviewed papers with Andrew Morin (Idaho National Labs) at Pre-ICIS/WISP 2025 and WOSOC/NDSS Symposium 2025; presented research at TU Research Day 2026.

Acting Department Chairperson and Lecturer | *Harare Institute of Technology* 2012 - 2023
Zimbabwe | Department of Computer Science, progression from Teaching Assistant through Chairperson

- Developed and delivered course materials for 18 subjects, including AI and Expert Systems, Neural Networks and Pattern Matching, Smartcard Cryptosystems, and Mobile Computing and Wireless Networks; supervised 2 MTech and 15 BTech research projects.
- As chair managing 10 faculty, launched 3 new degree programs (1 BTech, 2 MTech), digitized approximately 70 percent of departmental SOPs, and established an Industry Advisory Board of 8 experts.

KEY PROJECTS

Sensei: Multi-Agent AI Coaching Platform | *Hacklahoma 2026, Feb 2026*

- Built and deployed a multi-agent AI coaching app in 22 hours as part of a 4-person team, using Claude Sonnet API (Coach and Reflection agents with tool-use loops), TanStack Start, Cloudflare Workers, D1/SQLite, Drizzle ORM, Google Calendar API, and Google OAuth. Devpost: devpost.com/software/project-sensei

SOC Strategic Alignment Simulator and GCM Explorer | *PhD Research, 2024-Present*

- Research simulation of the SOC Speed-Quality Paradox; interactive Goal-Coherence Matrix for visualizing CISO-analyst goal misalignment; LLM Agreement Dashboard for cross-model validation. Live: wellytg.github.io/soc-paradox-sim

DigiVault: Automated OSINT and Breach Analysis | *Hurricane Hackathon 2026*

- Automated public profile construction and breach signal detection: real-time OSINT aggregation, exposure scoring, AI-powered recommendations via Claude API. Live: wellytg.github.io/digivault

EDUCATION

Doctor of Philosophy, Cyber Studies | The University of Tulsa, Tulsa OK Expected May 2027
SOC goal incongruency, quantitative security metrics design. Adviser: Tyler Moore.

Master of Technology, Computer Science | JNTUH College of Engineering Hyderabad
Bachelor of Technology, Computer Science | Harare Institute of Technology

PUBLICATIONS

Measuring Strategy Coherence in Security Operations Centers: A Goal-Metric Mapping Framework, with Andrew Morin | Pre-ICIS/WISP 2025 | aisel.aisnet.org/wisp2025/15

Mismatched SOCs: A Case Study of Cybersecurity Goal Incongruency Within An Academic SOC, with Andrew Morin | WOSOC 2025, co-located NDSS Symposium, San Diego

CERTIFICATIONS AND TRAINING

- ISC2 Certified in Cybersecurity (CC), active through Aug 2028
- Google Cybersecurity Professional Certificate V2 (2024; updated curriculum completed Jul 2026)
- Google AI Professional Certificate, 7-course specialization (Jul 2026)
- Cribl Certified Admin - Stream (Level 2); Cribl Certified User (Level 1)
- AI for App Deployment, Google (Aug 2026)
- Building LLM Applications With Prompt Engineering, NVIDIA Deep Learning Institute (Sep 2026)
- AWS Academy Graduate - Cloud Foundations - Training Badge (2022)